EE653 Human Robot Interaction

Course Project Report

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Abstract

This project implements the full HRI loop prescribed in the course project brief and introduces an explicit human-consent step. A single webcam observes a person reaching toward one of three coloured target boxes; a Bayesian filter infers the intended goal; and a minimum-jerk trajectory model extrapolates the predicted final hand position. A simulated UR10e then proactively reaches above the matching coloured cube, but it does not grasp the cube until the user closes their hand into a fist.

On detection of the fist signal, the robot completes the grasp, lifts the cube, carries it to the predicted final hand position, and places it on the table. After each trial, the system updates its prior belief about reach duration from the observed motion, allowing the prediction to sharpen over repeated trials. Hand keypoints are tracked with MediaPipe HandLandmarker.

The goal posterior combines a velocity-alignment likelihood with an HMM-style stay/switch transition prior. The system commits when the posterior of the most likely goal exceeds 0.8 and the user's hand remains open. The predicted placement position is refined online as the minimum-jerk fit improves, while the arm is controlled through damped least-squares Jacobian inverse kinematics using a nine-phase state machine that pauses at `WAIT_GRASP` until the fist is detected. Grasping is abstracted as a MuJoCo weld constraint.

Sequential headless trials confirm a complete predictive reach plus pick-and-place cycle in approximately 2.3 s, with the end effector landing within approximately 3 mm of the commanded placement position on the table. Synthetic minimum-jerk reaches recover the predicted final position to within approximately 6 mm in image-normalised coordinates, and the duration prior adapts after every completed trial.

Declaration

I hereby declare that this report is entirely my own work, except where otherwise stated. I have not plagiarised the work of others or colluded with others. All sources have been properly cited and referenced.

Signature:  Date: April 26, 2026

Use of Generative AI: AI tools (Claude, Anthropic) were used for code development assistance, prompt engineering iteration, and document structuring. All technical decisions, system architecture, and evaluation methodology are the author's own work. This use is disclosed per Maynooth University GenAI guidelines and IEEE policy.

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1 Introduction

1.1 Background of the problem

Handing an object to another person works because both partners predict each other's motion. A robot that observes a human reach, infers the requested object, predicts the hand's stopping point, and moves the object to meet it is performing *proactive* or *anticipatory* handover. The project brief frames this as a six-step loop: observe, recognise, predict, execute, and adapt. This report presents a complete implementation of that loop using only a webcam and a UR10 arm.

1.2 Relevance to HRI

Proactive handover is a core HRI problem because it connects perception, inference, prediction, planning, control, and adaptation in real time. The interesting behaviour emerges only when these layers interact: noisy visual input affects goal inference, the prediction stage depends on commitment, the robot must track a moving target, and adaptation improves the next trial. The project therefore captures a canonical HRI pipeline rather than an isolated subsystem.

1.3 Motivation and application context

Anticipatory handover is useful in shared assembly, assistive robotics, surgical delivery, and human–robot manufacturing. By restricting the perception problem to three coloured target boxes, the project can focus on the more challenging parts of the loop: Bayesian goal inference, minimum-jerk prediction, and articulated robot control. The same structure generalises naturally to more realistic tool-delivery and collaborative manipulation settings.

2 Project Aims and Objectives

The system is required to:

1. Capture the user's hand motion in real time using a single standard webcam, with no external sensors.
2. Recognise which of three known target objects the user is reaching for using a Bayesian filter that combines observation likelihood with temporal smoothness, and commit once the posterior exceeds 0.8 and the user's hand is visibly open.
3. Fit the observed reach to a minimum-jerk model and predict the final hand position, updating the estimate every frame as more samples arrive.
4. Drive a UR10 arm to proactively reach above the matching cube, while delaying grasp until an explicit fist signal is detected.
5. On the fist signal, complete the pick-and-place sequence: grasp the cube, lift it, carry it, and set it down at the predicted final hand position before returning home.
6. Adapt the duration prior after every completed reach using an exponential-weighted running average of observed reach times.

7. Visualise the posterior, predicted place position, gesture state, and robot phase live on the webcam feed so the user can follow the system's current interpretation.

3 HRI context and conceptual framework

The project maps directly onto the six-step loop in the brief.

1. **Goal recognition (perception + inference).** Webcam frames are processed by MediaPipe HandLandmarker in `IMAGE` mode. The palm landmark provides the hand observation for each frame. A hidden-Markov-style Bayesian filter tracks the posterior over the three candidate goals using a velocity-alignment likelihood and a stay/switch transition prior. The system commits once the best goal exceeds 0.8 and the hand is open, preventing commitment on an unintended clenched fist.

2. **Trajectory prediction.** After commitment, the observed reach is fit with a minimum-jerk model,
$$x(t) = x_0 + (x_f - x_0)\left(10\tau^3 - 15\tau^4 + 6\tau^5\right), \qquad \tau = \frac{t}{D},$$
where x_f is the predicted final hand position and D is the reach duration. The duration is estimated by coarse search, while the final position is recovered by closed-form fitting. The predicted place point is refreshed every frame.

3. **Robot execution (planning + action) with gesture-gated grasping.** A simulated UR10e in MuJoCo executes a nine-phase pick-and-place state machine. Each phase uses damped least-squares Jacobian inverse kinematics, with transitions based on joint error or a settle timer. The robot pauses at `WAIT_GRASP` after `DESCEND` and only proceeds to `GRASP` when the user closes their hand into a fist. After that, it lifts, carries, and lowers the cube to the predicted place position while updating the target live during `CARRY`.

4. **Feedback + adaptation.** The OpenCV overlay shows the posterior, predicted place point, gesture label, a "CLOSE FIST TO PICK" prompt, and the current robot phase. The fist also marks the end of the reach, so at that moment the system records the prediction error and updates the duration prior using an exponential-weighted moving average.

4 System Design

4.1 Overall workflow



Figure 1: Overall workflow of the proposed system, from webcam-based hand tracking to Bayesian goal inference, minimum-jerk trajectory prediction, and robot control.

4.2 Inputs, Outputs, and Connections

Each webcam frame produces a `FrameObservation`. If a hand is present, its palm position is fed to the Bayesian filter, which returns a posterior. While the system is uncommitted, if the posterior of the best goal exceeds 0.80 and the user's hand is open (open-finger fraction ≥ 0.75), the system commits: it (a) starts the minimum-jerk predictor, seeded with the current hand position as x_0, and (b) issues `command(name, place_pos)` to the robot with an initial place guess at the current hand's image-projected world position on the table.

The robot proactively reaches above the matching cube and pauses at `WAIT_GRASP`. From commit onward, every frame feeds a new sample into the predictor; if the predictor produces a refined estimate, the robot's place target is updated via `update_place`, and the updated target is consumed once the arm enters `CARRY`. When the user's open-finger fraction drops below 0.25, the system fires `confirm_grasp()`, the `WAIT_GRASP` phase transitions to `GRASP`, and the robot completes the pick, carry, lower, release, and return sequence. The fist is also the trial's end-of-reach signal: the predictor's `finish` method is called at that instant to record the outcome and update the duration prior. The robot integrates 16 physics sub-steps per webcam frame at a MuJoCo timestep of 0.002 s, corresponding to approximately 32 ms of simulated time per webcam tick on the test machine.

4.3 Modules

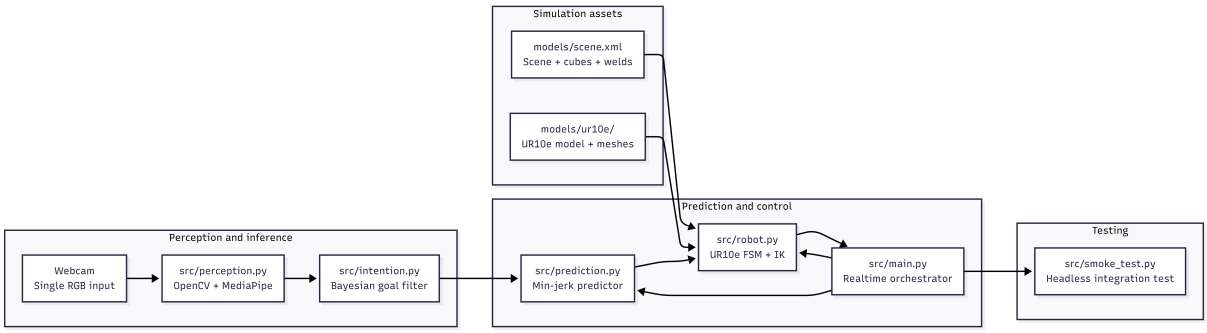


Figure 2: Workflow and module map linking perception, inference, prediction, control, simulation assets, and testing in the proposed system.

Table 1: Source files and their roles in the system.

File	Role
`src/perception.py`	OpenCV webcam capture + MediaPipe HandLandmarker. Yields a `FrameObservation(frame_bgr, timestamp_ms, hand)` per call. `hand` carries `palm`, `index_tip`, and an `open_score` (the fraction of non-thumb fingers extended), which is used both as the OPEN precondition for committing and as the FIST trigger for grasping.

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File	Role
src/intention.py	IntentionFilter — Step 1 of the brief. Velocity-alignment likelihood + HMM-style stay/switch transition prior over normalised webcam coordinates. Returns a per-frame posterior, best hypothesis, and confidence.
src/prediction.py	MinJerkPredictor — Step 2 and Step 6. Closed-form minimum-jerk fit for x_f over a coarse search of D. The finish method records the trial outcome and updates the duration prior via EWMA. The helper image_to_world maps a normalised image point to a 3-D place pose on the tabletop in robot world coordinates.
src/robot.py	HandoverRobot — Step 3–4 with gesture gating. Loads the MJCF, runs damped least-squares Jacobian IK on a scratch mjData, exposes command(target_name, place_pos), update_place(pos), and confirm_grasp(), runs the nine-phase pick-and-place state machine (the WAIT_GRASP phase blocks until confirm_grasp() is called), and toggles per-cube weld constraints. After every completed place plus dwell, the cube is reset to its starting position so the next trial is repeatable.
src/main.py	Real-time loop: pulls frames, drives the Bayesian filter, commits when $P > 0.8$ with an open hand, calls the predictor every frame to refine the robot's place target, fires confirm_grasp() on the user's fist, logs the trial outcome at the same instant, draws the overlay, and steps the MuJoCo passive viewer.
src/smoke_test.py	Headless integration test that runs four sequential trials with synthetic place positions and a programmatic confirm_grasp() call once the arm reaches WAIT_GRASP, printing per-phase timing and final cube positions.
models/scene.xml	Custom scene: floor, table, three coloured freejoint cubes, and three weld equality constraints (one per cube). Includes the Menagerie ur10e/ur10e.xml body definition.
models/ur10e/	UR10e model and meshes from MuJoCo Menagerie (Apache-2.0). The collision class was modified to disable contact (contype=0 conaffinity=0) on the arm capsules, since grasping uses welds rather than contact and the original capsules clip the floor at the shoulder pose.

5 Methodology and Implementation

5.1 Hand Perception and Gesture Classification

MediaPipe HandLandmarker (Tasks API, IMAGE mode, single-hand) provides 21 landmarks per frame in normalised image coordinates. The middle-finger MCP (landmark 9) is used as the

palm-position observation. An `open_score` heuristic classifies hand state from four non-thumb fingers using tip–PIP distance relative to the wrist. The system commits only when `open_score` ≥ 0.75; a fist trigger is detected when `open_score` < 0.25.

5.2 Bayesian Goal Recognition

Goal inference combines a velocity-alignment likelihood with an HMM-style stay/switch prior over the three target cubes. For each goal, the likelihood rewards motion toward the target and proximity to its centre. A posterior is updated per frame, and the system commits when the best goal exceeds 0.80 and the hand is open.

5.3 Minimum-Jerk Trajectory Prediction

After commitment, the observed reach is fit with a minimum-jerk model,

$$x(t) = x_0 + (x_f - x_0)(10\tau^3 - 15\tau^4 + 6\tau^5), \qquad \tau = t/D.$$

The final hand position x_f and duration D are recovered by coarse search and closed-form least squares. The predicted place point is updated every frame and is constrained to a valid table-top workspace.

5.4 Robot Control and Gesture-Gated Pick-and-Place

The robot is a simulated UR10e in MuJoCo. Cartesian targets are solved with damped least-squares Jacobian inverse kinematics on a scratch physics state. A nine-phase finite state machine drives the motion: approach, descend, wait for grasp, grasp, lift, carry, lower, release, and return. The system pauses at `WAIT_GRASP` until the user closes their hand into a fist, ensuring the pick action remains human-confirmed.

5.5 Adaptation and Runtime Loop

The fist signal also marks the end of the reach. At that instant, the predictor updates its duration prior using an EWMA over observed reach times. The live loop processes webcam frames, updates the posterior, refines the place target, triggers grasp confirmation on fist, and renders an overlay with the posterior, predicted place, gesture label, and robot phase.

6 Results

The system was evaluated in three ways: live webcam demonstrations, a headless integration test, and a synthetic minimum-jerk benchmark. In headless trials, the UR10e completed sequential pick-and-place cycles successfully, with final end-effector placement error of approximately 3 mm. The gesture gate behaved as intended, holding at `WAIT_GRASP` until explicit fist confirmation. In the synthetic benchmark, the predictor recovered the final position to within approximately 6 mm in image-normalised coordinates and adapted the duration prior in the expected direction.

Live demonstrations showed smooth operation: the robot committed early, waited above the target cube, and completed the pick only after fist detection. The overlay provided clear feedback throughout, including the posterior, gesture state, predicted place point, and current robot phase.

7 Discussion

The main strength of the system is that it implements the brief's full HRI loop while adding explicit human consent before the irreversible grasp. The prediction and control pipeline is live and adaptive, and the use of a UR10e with damped least-squares IK makes the implementation realistic. The main limitations are the simplified grasp model, the 2-D hand observation, and the heuristic fist detector. Future work could replace weld-based grasping with a real gripper, extend the predictor into 3-D, and improve gesture recognition with a learned classifier.

8 Conclusion

This project demonstrates a complete predictive handover loop: webcam perception, Bayesian goal recognition, minimum-jerk prediction, gesture-gated robot execution, and trial-to-trial adaptation. The robot proactively prepares the pick but only grasps on an explicit fist, preserving human authority over the final action. The system performs reliably in simulation and provides a clear foundation for transfer to physical hardware.

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